

Lab06 - Computer Vision

MTRN3100 - UNSW School of Mechanical and Manufacturing Engineering

Prerequisites

This lab uses Python. If not yet installed, please follow the instructions on Moodle.

Task 1 - Load An Image

Load in the provided “shapes.png” image and display it using matplotlib. Ensure you display it in the correct RGB format and not BGR.

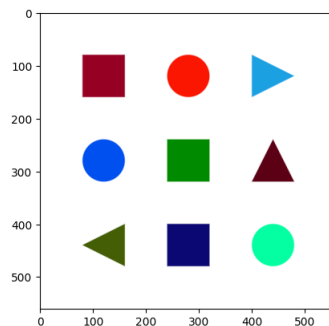


Figure 1: Example of the image displayed in the correct format for Task 1.

Questions

- **Question 1:** How is a true colour image (24-bit RGB) represented in memory?
- **Question 2:** What is the difference between RGB and BGR image representations in terms of colour channel ordering?

Task 2 - Colour Mask

Using a blue colour mask, generate a monochrome mask of the blue shapes.

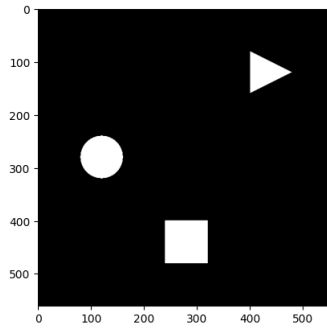


Figure 2: Example of the blue mask in Task 2.

Questions

- **Question 3:** Explain how a blue colour mask isolates specific regions of an image. Describe the role of colour thresholds for this task.

Task 3 - Bitwise Operation

Using the bitwise and operation, apply the mask to the source image and display the result.

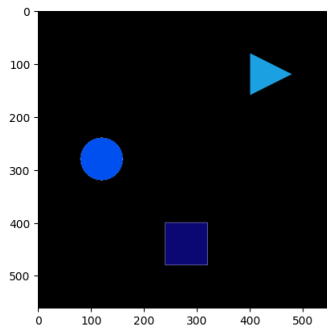


Figure 3: Example of a bitwise operation with the mask and source image.

Task 4 - HSV and Maze

HSV (Hue, Saturation, Value) may provide better results for the Micromouse maze. Use the provided `hsv_picker.ipynb` to find a good range of HSV values for the maze and display the resulting mask of the maze walls.

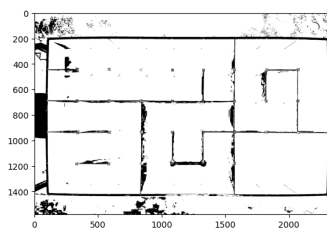


Figure 4: Example of an HSV mask. Note: your mask does not have to match this exactly.

Questions

- **Question 4:** Explain why it may be better to represent an image in HSV format rather than RGB for certain image processing tasks.

Task 5 - Erode

Perform an erode operation on the source image. Use a kernel size of 7 and perform 5 iterations.



Figure 5: Example erode performed on the mask shown in Task 4.

Questions

- **Question 5:** In terms of image processing, what is a kernel?
- **Question 6:** In terms of image processing, how is a kernel typically applied to an image?
- **Question 7:** Describe what the erosion operation does to an image mask. What types of applications might erosion be useful?

Task 6 - Dilate

Perform a dilate operation on the source image. Use a kernel size of 3 and perform 5 iterations.

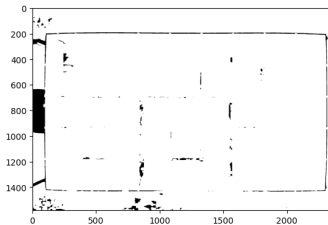


Figure 6: Example dilate performed on the mask shown in Task 4.

Questions

- **Question 8:** Explain how the dilation operation affects the binary mask of an image. What types of applications might dilation be useful?
- **Question 9:** Why might it be preferable to run a smaller kernel over the image for multiple iterations rather than using a much larger kernel in a single operation?
- **Question 10:** How does the size of the kernel affect the result of erosion or dilation operations on a binary mask?

Extension 1 - ASCII Representation

Note: This task should be completed as it is part of the Micromouse assignment

Convert the maze into an ASCII representation shown in the lab document. You may need to perform a series of erosions and dilations to get a good mask.

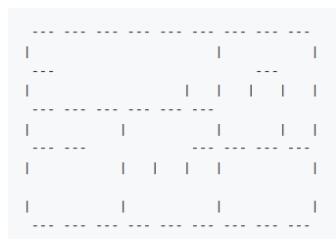


Figure 7: Example ASCII representation of the provided maze.